

PC603CS	DISTRIBUTED SYSTEMS				
Prerequisites	Operating Systems	L	T	P	C
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Evaluation	CIE	40 Marks	SEE	60 Marks	

Course Objectives :	
1.	To acquire an understanding of the issues in distributed systems.
2.	To study architectures and working of distributed file systems.
3.	To expose the students to distributed transaction management, security issues and replication.
4.	To explore knowledge on Distributed fault tolerant Systems
5.	To understand map-reduce algorithms.

Course Outcomes : At the end of the course the student will be able to:	
1.	Describe the problems and challenges associated with distributed systems.
2.	Implement small scale distributed systems.
3.	Understand design tradeoffs in large-scale distributed systems.
4.	Gain knowledge on Consistency and Replication
5.	Familiar with Distributed Fault Tolerant systems

UNIT – I
Introduction: From networked systems to distributed systems , Design goals, A simple classification of distributed systems , Pitfalls Architectures: Architectural Styles, Middleware and distributed systems, Layered-system architectures, Symmetrically distributed system architectures , Hybrid system architectures Processes: Threads, Virtualization, Clients, Servers, and Code Migration. Communication: Foundations, Remote Procedure Call, Message-Oriented Communication, Stream-Oriented Communication, and Multicast Communication.

UNIT – II
Coordination : Clock Synchronization, Logical Clocks, Mutual Exclusion, Election Algorithms, Gossip based Coordination, Distributed Event Matching, Location Systems. Transactions- The Slippery Concept of a Transaction, Weak Isolation Levels, serializability Naming: Names, Identifiers and Addresses, Flat Naming, Structured Naming, and Attribute-Based Naming, Named Data Networking

UNIT – III

Consistency and Replication: Introduction, Data-Centric Consistency Models, Client-Centric Consistency Models, Replica Management, and Consistency Protocols.

Replication- Leaders & Followers, Problems with Replication Lag, Multi-Leader Replication, Leaderless Replication

Partitioning - Partitioning and Replication, Partitioning of Key- Value Data, Partitioning and Secondary Indexes, Rebalancing Partitions, Request Routing

Consistency And Consensus- Consistency Guarantees, Linearizability, Ordering Guarantees, Distributed Transactions and Consensus

UNIT – IV

The Trouble with Distributed Systems- Faults and Partial Failures, Unreliable Networks, Unreliable Clocks, Knowledge, Truth and Lies

Fault Tolerance: Introduction to Fault Tolerance, Process Resilience, Reliable Client-Server Communication, Reliable Group Communication, Distributed Commit, and Recovery.

UNIT – V

Map-Reduce and Distributed File Systems : Example, Scaling, programming model, Apache Hadoop, Amazon Elastic Map Reduce, Mapreduce.net, Pig and Hive.

Beyond Map Reduce

References:

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| 1. | Maarten Van Steen, and Andrew S. Tanenbaum, <i>Distributed Systems</i> , PHI 4 nd Edition, 2023 |
| 2. | Martin Kleppman , <i>Designing Data Intensive Systems</i> , O'Reilly, 2017 |

Suggested Readings:

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| 1. | R.Hill, L.Hirsch, P.Lake, S.Moshiri, <i>Guide to Cloud Computing, Principles and Practice</i> , Springer, 2013. |
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